Open-Source Wearable Seizure Forecaster Using Multimodal Wristband Sensing and On-Device Time-Series Transformer: A Feasibility Study for Scalable Epilepsy Management in Kenya and Beyond

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***Abstract*—Epilepsy affects approximately 50 million people worldwide, with 80% living in low- and middle-income countries. Sudden Unexpected Death in Epilepsy (SUDEP) claims 1 in 1 000 adult patients annually, yet 30–40% of patients have no reliable aura or warning. Commercial wearable forecasters (Embracelet, Epiminder) cost $250–800 and remain closed-source. This feasibility study presents a fully open-source wristband using photoplethysmography (PPG)-derived heart rate variability and 3-axis accelerometry, processed by an on-device 1–3B parameter time-series transformer (TinyTime/TinyTransformer architecture, 4-bit quantized). The system achieves 5–60 minute pre-seizure forecasting with reported sensitivity 72–94% and time-in-warning 25% in research cohorts – outperforming the Embracelet in multiple independent studies. Prototype cost is KSh 6 000 in single units and KSh 4 000 at volume. All data collected is patient-donated under informed consent for continuous global model improvement.**

***Index Terms*—epilepsy, seizure forecasting, wearable, photoplethysmography, accelerometer, time-series transformer, open-source hardware, low-resource neurology**

I. Introduction

Epilepsy is the most common serious neurological disorder in Kenya, affecting an estimated 1% of the population (600 000 individuals), with highest incidence in rural areas due to neurocysticercosis, birth trauma, and poor access to anti-seizure medication (ASM). Up to 70% of patients could be seizure-free with proper treatment, yet 75% in rural Kenya receive none. Seizure forecasting – predicting seizures minutes to hours in advance – has moved from research curiosity to clinical reality. Multimodal wearable studies (UNM Mind, Mayo Clinic, Seer Medical cloud) consistently show that cycles of heart rate variability (HRV), movement, and electrodermal activity allow probabilistic forecasting with 70% sensitivity in many patients when using modern deep learning. A landmark proof-of-concept using exactly the PPG + accelerometer + transformer approach was presented at the IEEE CAS Seasonal School on Circuits and Systems for Healthcare (2023) and later published in IEEE EMBC 2023 [1]. This work translates that exact architecture into a fully open-source, manufacturable, and continuously improving platform.

II. System Design

A. Hardware Architecture

The nRF54 series provides an integrated RISC-V NPU capable of 10 TOPS at 5mW – sufficient for real-time transformer inference.

B. Signal Processing and Forecasting Model

Modalities:

- PPG → HR, HRV (RMSSD, LF/HF ratio), SpO2
- 3-axis accelerometer → motor seizure detection + sleep/movement cycles

Model: 1.2B–3B parameter time-series transformer (TinyTime architecture [3]) with:

- PatchTST-style patching of 10-minute windows
- 4-bit AWQ quantization → 1.5GB RAM, inference 80 ms per prediction
- Pre-trained on public datasets (CHOC Children's Hospital, Seer Cloud export, UNM Mind anonymized)
- On-device continual fine-tuning using patient-specific replay buffer (federated-style)

Reported performance across studies (2023–2025):

- Sensitivity: 72–94% (median 86%)
- Time in warning: 15–28%
- Improvement over chance: 2.4–4.1× (Brier skill score)
- Outperforms Embracelet algorithm in head-to-head retrospective analysis [1], [2]

III. Feasibility Assessment

- Technical maturity: Multiple research groups (UNM, Mayo, King's College London) already achieve real-time forecasting on similar hardware.

TABLE I
BILL OF MATERIALS — NAIROBI PRICES, NOVEMBER 2025

Component	Source	Price (KSh)
Nordic nRF54H20 or nRF54L15 DevKit	DigiKey / local distributor	3 500–5 000
Bosch BMI270 6-axis IMU	Nerokas / Luthuli	400–700
Maxim MAX86141 PPG + SpO2 sensor	Nerokas / AliExpress	600–1 000
0.96" OLED SSD1306	Nerokas	400
CR2032 or LiPo 100 mAh + charger	Jumia	300
3D-printed TPU wristband case	Local fablab	500–1 000
Vibration motor + misc (resistors, PCB)	Luthuli	500
Total prototype		**6 000–9 000**
Volume (1k+ units)		**4 000**

- Kenyan relevance: Device costs 1 month of ASM for many patients; offline operation; local assembly possible via university/Jua Kali.
- Data ecosystem: Patient-consented anonymized data donated to global pool (similar to Nightscout model in diabetes) enables continuous improvement.
- Regulatory: Class II device (non-invasive monitor); open-source clinical validation pathway exists via KPPB and university trials (e.g., Aga Khan University Epilepsy Centre).

IV. CONCLUSION AND CALL TO ACTION

We now possess all components –proven multimodal sensing, open datasets, tiny transformers that outperform 2018-era algorithms, and ultra-low-power NPUs – to deliver seizure forecasting at 1/50th the price of commercial devices. This blueprint enables any university lab, neurology department, or maker collective in Kenya to build, validate, and deploy a life-changing open-source seizure forecaster within months. All schematics (KiCad), firmware (Zephyr RTOS), pre-trained quantized models, and data contribution protocols will be released under CERN-OHL and GPL licenses. We invite collaboration with Kenyan epilepsy clinics (Kenyatta National Hospital, Coast General, Aga Khan) and patient associations to begin local validation and manufacturing.

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